

FBISE Computer Science Class 9

1st Annual Examination 2026

Complete Solutions of Three Question Papers

Papers B, D and M | MCQs, Section B, Section C and all OR
alternatives

Paper B - Complete Solution

SECTION A - MCQs (12 marks)

Choose the option shown below. The reason is included for conceptual understanding.

1. Which document outlines a business idea, its goals, and planning details?

Correct Answer: B — Business plan

Reason: A business plan describes a business idea, objectives, target customers, financial planning and strategies.

Why other options are wrong:

- **A — Resume:** Describes a person's education, skills and work experience.
 - **C — Invoice:** Shows details of products/services purchased and the amount to be paid.
 - **D — Advertisement:** Promotes or informs people about a product, service or event.
-

2. Why is computational thinking important in computer science?

Correct Answer: B — It helps break problems into manageable steps

Reason: Computational thinking divides a complex problem into smaller and easier parts.

Why other options are wrong:

- **A — It increases typing speed:** Computational thinking is about problem-solving, not typing speed.
 - **C — It improves computer hardware:** Hardware improvement requires engineering and physical components.
 - **D — It replaces programming:** Computational thinking helps us design programs; it does not replace programming.
-

3. Why is HTML used when creating a website?

Correct Answer: C — To display text and images

Reason: HTML creates the structure and content of a webpage, including headings, paragraphs, images and links.

Why other options are wrong:

- **A — To style web pages:** CSS is mainly used for colors, fonts, spacing and layout.

- **B — To store data:** Databases and storage systems are used to store data.
 - **D — To run programs:** Programming languages and scripting languages execute programs. HTML only defines webpage structure.
-

4. Which part of development handles the user interface of a website?

Correct Answer: D — Front end

Reason: The front end is the visible part of a website that users see and interact with. It uses HTML, CSS and JavaScript.

Why other options are wrong:

- **A — Back end:** Handles server-side logic, databases and application processing.
 - **B — Server side:** Refers to operations performed on the web server, not the visible interface.
 - **C — Database layer:** Stores and manages data; it does not create the user interface.
-

5. Why are variables used in JavaScript?

Correct Answer: A — To store values that can change

Reason: A variable stores data that can be used and updated during program execution.

Example:

```
let marks = 75;  
marks = 85;
```

The value of marks changes from 75 to 85.

Why other options are wrong:

- **B — To design web pages:** HTML provides structure, while CSS is used for webpage design.
- **C — To create hyperlinks:** HTML uses the <a> tag to create hyperlinks.
- **D — To format text:** HTML and CSS are used for text formatting.

6. Why is the ALU an important part of the CPU?

Correct Answer: A — It performs arithmetic and logical operations

Reason: The Arithmetic Logic Unit (ALU) performs calculations such as addition, subtraction, multiplication and division. It also performs logical comparisons such as greater than, less than and equal to.

Why other options are wrong:

- **B — It manages memory:** Memory management is mainly handled by the control unit and operating system.
 - **C — It controls input devices:** Input devices are controlled through the input/output system and control unit.
 - **D — It performs analogue linear operations:** Computers mainly process digital data, while the ALU performs digital arithmetic and logic operations.
-

7. Why is cloud computing needed in modern computing?

Correct Answer: C — To access data and services over the Internet

Reason: Cloud computing provides online access to storage, software, databases and processing services from anywhere through the Internet.

Why other options are wrong:

- **A — To reduce screen size:** Screen size is a hardware design feature and is unrelated to cloud computing.
 - **B — To replace computers:** Cloud computing provides services through the Internet; it does not completely replace computers.
 - **D — To increase keyboard speed:** Keyboard speed depends on the user and keyboard hardware, not cloud computing.
-

8. What is meant by fake news?

Correct Answer: C — False or misleading information

Reason: Fake news is false, inaccurate or misleading information presented as genuine news.

Why other options are wrong:

- **A — Verified information:** Verified information has been checked and confirmed.
 - **B — Educational content:** Educational content is prepared to teach or explain a topic.
 - **D — Official announcements:** Official announcements are issued by authorized organizations or institutions.
-

9. What do Sustainable Development Goals (SDGs) represent?

Correct Answer: D — Global development goals

Reason: SDGs are worldwide goals established to improve society, the economy and the environment. They include quality education, good health, clean water and poverty reduction.

Why other options are wrong:

- **A — Business profits:** Business profits are financial earnings, not global development goals.
 - **B — Personal goals:** Personal goals are targets set by individuals.
 - **C — Marketing targets:** Marketing targets are related to selling and promoting products.
-

10. How does logical thinking support problem-solving?

Correct Answer: B — By arranging steps in a correct sequence

Reason: Logical thinking helps organize facts, decisions and actions in a clear and correct order to reach the desired result.

Why other options are wrong:

- **A — By using trial and error and providing false results:** Logical thinking aims to produce correct and reliable results.
 - **C — By memorizing commands and repeating these:** Problem-solving requires reasoning, not only memorization.
 - **D — By designing interfaces:** Interface design is a part of web or software development, not the main purpose of logical thinking.
-

11. Why is software piracy considered illegal?

Correct Answer: A — It violates ownership rights

Reason: Software is protected by copyright. Copying, distributing or using software without the owner's permission violates intellectual-property and ownership rights.

Why other options are wrong:

- **B — It is outdated:** Software piracy is illegal because it violates copyright, not because the software is outdated.
- **C — It slows the working of the computer:** Pirated software may contain malware, but this is not the legal reason for piracy being illegal.
- **D — It limits access to online data:** Limiting online data access is unrelated to software piracy.

12. Why is continuous data suitable for measurements like the heights of students?

Correct Answer: D — It can take any value in a range

Reason: Continuous data can have decimal or fractional values within a range. A student's height may be 150 cm, 150.5 cm, or 150.55 cm. Therefore, height is continuous data.

Why other options are wrong:

- **A — It uses labels:** Labels describe categorical data, such as blood group, colour or gender.
- **B — It has fixed values:** Fixed values are not suitable for measurements that can vary continuously.
- **C — It is non-numeric:** Height is numerical data because it is measured using numbers.

Section B — Short Questions

Q.2 (i)

(a) What is the main function of the Arithmetic Logic Unit (ALU)? Explain briefly.

The **Arithmetic Logic Unit (ALU)** is a major component of the CPU. Its main function is to perform:

1. **Arithmetic operations**, such as addition, subtraction, multiplication and division.
2. **Logical operations**, such as AND, OR and NOT.
3. **Comparison operations**, such as checking whether one value is greater than, less than or equal to another value.

Thus, the ALU performs the mathematical calculations and logical decision-making required while executing instructions.

OR: What is meant by problem decomposition? Elaborate briefly.

Problem decomposition is the process of breaking a large and complex problem into smaller, simpler and manageable sub-problems. Each sub-problem can be understood, solved and tested separately. The solutions of all the smaller parts are then combined to solve the complete problem.

Example: An online shopping system can be decomposed into user registration, product searching, shopping cart, payment and order-delivery modules.

Q.2 (ii)

(a) Why is computational thinking important in computer science? Write briefly.

Computational thinking is important because it provides a systematic way to understand and solve complex problems. It involves:

- **Decomposition:** Breaking a problem into smaller parts.
- **Pattern recognition:** Identifying similarities or repeated patterns.
- **Abstraction:** Focusing on relevant information and ignoring unnecessary details.
- **Algorithm design:** Developing step-by-step instructions for solving the problem.

It helps programmers design efficient algorithms, write organised programs and develop solutions that can be executed by computers.

OR: How does CSS improve the appearance of a web page? Give an example.

CSS (Cascading Style Sheets) improves the appearance of a web page by controlling its colors, fonts, spacing, borders, alignment and overall layout. It separates the presentation of a web page from its HTML content, making the page more attractive, consistent and easier to maintain.

Example:

```
<p class="message">Welcome to our website!</p>
.message {
    color: blue;
    font-size: 20px;
    font-weight: bold;
}
```

This CSS displays the paragraph in **blue**, with a **20-pixel font size** and **bold text**.

Q.2 (iii)

(a) State two functions of an operating system.

1. Process management:

The operating system manages all running programs and allocates CPU time to them. It also supports multitasking by allowing several programs to run efficiently.

2. Memory management:

It allocates and deallocates main memory to different programs. It also ensures that one program does not improperly access the memory assigned to another program.

Other valid functions include file management, device management, security and providing a user interface.

OR: What is meant by phishing? Give one example.

Phishing is a cyberattack in which an attacker pretends to be a trusted person or organisation to deceive users into revealing sensitive information, such as passwords, bank-account details or credit-card numbers. It is commonly carried out through fake emails, messages or websites.

Example: A user receives a fake email that appears to be from a bank. The email asks the user to click a link and enter their login details. The linked website is fake and sends the entered information to the attacker.

Q.2 (iv)

(a) Write down two advantages of Big Data.

1. Better decision-making:

Big Data allows organizations to analyze large amounts of information, identify trends and make faster, evidence-based decisions.

2. Improved services and personalization:

Organizations can study customers' preferences and behavior to provide personalized recommendations, targeted services and a better customer experience.

Example: An online shopping platform analyses customers' searches and purchases to recommend relevant products.

OR: A computer is running slowly while many programs are open. Which hardware component is mainly responsible for processing instructions? Also give a reason.

The **Central Processing Unit (CPU)** is mainly responsible for processing and executing program instructions.

When many programs are open, the CPU must divide its processing time among multiple programs and repeatedly switch between their tasks. This increases the CPU workload and may slow down the computer, particularly if the CPU has limited speed or fewer processing cores.

However, insufficient **RAM** can also contribute to the slowdown because the computer may have to transfer data between RAM and slower secondary storage. Still, the hardware component that actually **processes instructions** is the CPU.

Q.2 (v)

(a) Write one harmful effect of using computing innovation.

One harmful effect of computing innovation is the **loss of privacy**. Modern applications, websites and smart devices collect large amounts of users' personal information, including their locations, interests, online activities and financial details.

If this information is collected without proper consent, misused or accessed through a security breach, it may result in identity theft, financial loss or unauthorized surveillance. Therefore, computing innovations must include strong privacy and security measures.

OR: How does Data Science differ from Data Analytics?

Data Science

Data Science is a broad field that uses programming, statistics, machine learning and data analysis to extract knowledge from data.

It may work with structured as well as unstructured data and can build predictive models.

It often predicts what may happen in the future.

Data Analytics

Data Analytics mainly examines existing data to identify patterns and draw useful conclusions.

It usually works with organized historical data to answer specific questions.

It primarily explains what happened and why it happened.

Example: A data scientist may develop a machine-learning model to predict future sales, whereas a data analyst may examine previous sales reports to determine which product sold the most.

Q.2 (vi)

(a) Correct any three errors in the given HTML code.

The tags in the given code are incorrectly ordered, and some opening/closing tags are missing or incorrect.

Correct HTML code:

```
<html>
<head>
  <title>Main Page</title>
</head>

<body>
  <u>Introduction to Computer</u>
</body>
</html>
```

Errors corrected:

1. <html> must be the opening tag, while </html> must appear at the end.
2. The title must be written inside the head section as <title>Main Page</title>.
3. The correct order is <head> ... </head>, not </head> ... <head>.
4. <body> must be written before the page content and </body> after it.
5. The underline opening tag is <u>, not .
6. The correct closing underline tag is </u>.

OR: How does packet switching transmit data over a computer network?

In **packet switching**, a large message is divided into small units called **packets**. Each packet contains part of the original data along with control information, such as the source address, destination address and sequence number.

The packets may travel through different routes across the network depending on traffic and route availability. When they reach the destination, they are arranged in the correct sequence and reassembled to form the original message. If a packet is lost or damaged, only that packet needs to be retransmitted.

Q.2 (vii)

(a) Differentiate between a dataset and a database.

Dataset	Database
A dataset is a collection of related data generally used for analysis, research or machine learning.	A database is an organised collection of data stored and managed electronically.
It is often stored in a single file, such as a CSV or spreadsheet file.	It may contain multiple related tables and is managed through a DBMS.
It is usually static and created for a specific purpose.	It is dynamic and supports adding, updating, deleting and retrieving data.
Example: A CSV file containing students' marks.	Example: A school database containing student, teacher and fee records.

OR: Write HTML code to apply headings from <h1> to <h6>.

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML Headings</title>
</head>

<body>
  <h1>Heading Level 1</h1>
  <h2>Heading Level 2</h2>
  <h3>Heading Level 3</h3>
  <h4>Heading Level 4</h4>
  <h5>Heading Level 5</h5>
  <h6>Heading Level 6</h6>
</body>
</html>
```

<h1> represents the largest and most important heading, while <h6> represents the smallest and least important heading.

Q.2 (viii)

(a) Why are variables used in JavaScript programs? Explain briefly. Give one example.

A **variable** is a named memory location used to store data in a program. Variables allow a JavaScript program to store, access and modify values during its execution. These values may include numbers, text or Boolean values.

Variables make programs flexible because the same variable can be used repeatedly, and its value can be changed when required.

Example:

```
let marks = 85;  
let bonus = 5;  
let totalMarks = marks + bonus;
```

```
console.log(totalMarks);
```

Here, marks, bonus and totalMarks are variables. The output is 90.

OR: Compare natural and artificial systems.

Natural systems	Artificial systems
Natural systems are created by nature without human intervention.	Artificial systems are designed and created by human beings.
They consist of naturally occurring components that interact with one another.	Their components are organised to achieve a particular human-defined purpose.
Examples include the solar system, ecosystem and human digestive system.	Examples include a computer system, transportation system and banking system.

Q.2 (ix)

(a) Why is identifying inputs, processes and outputs necessary?

Identifying **inputs, processes and outputs** is necessary because it:

1. **Clarifies requirements:** It determines what data must be entered into the system.
2. **Defines operations:** It explains how the input will be processed to solve the problem.
3. **Specifies results:** It identifies the output that the system is expected to produce.

This analysis helps programmers understand a problem clearly and design an accurate and efficient solution.

Example: In a program that calculates the area of a rectangle:

- **Input:** Length and width
- **Process:** Area = length × width
- **Output:** Area of the rectangle

OR: Write any three disadvantages of using Artificial Intelligence algorithms.

1. Bias and unfair decisions:

AI algorithms may produce biased results if they are trained using incomplete, inaccurate or biased data.

2. Lack of human judgement:

AI cannot completely understand emotions, ethics and complex human situations. Therefore, it may make technically correct but socially inappropriate decisions.

3. High cost and complexity:

Developing, training and maintaining AI systems may require expensive hardware, specialised software, large amounts of data and skilled professionals.

Other valid disadvantages include job displacement, privacy concerns, security risks and excessive dependence on technology.

Q.2 (x)

(a) What is meant by design thinking? Also write the names of its five phases.

Design thinking is a human-centred and iterative approach to problem-solving. It focuses on understanding users' needs, defining their problems and developing practical and innovative solutions. Solutions are improved continuously through testing and feedback.

The five phases of design thinking are:

1. **Empathize:** Understand the users, their needs and difficulties.
2. **Define:** Clearly state the main problem that needs to be solved.
3. **Ideate:** Generate different possible ideas and solutions.
4. **Prototype:** Develop a simple model or sample of the selected solution.
5. **Test:** Test the prototype with users and improve it using their feedback.

OR: Why is cloud computing needed for modern applications? Explain briefly.

Cloud computing provides computing resources—such as servers, storage, databases, software and processing power—through the Internet.

It is needed for modern applications because:

1. **Scalability:** Resources can be increased or decreased according to the number of users and application workload.
2. **Anywhere access:** Users can access applications and data from different devices and locations through the Internet.

3. **Cost-effectiveness:** Organisations do not need to purchase and maintain expensive local servers.
4. **Reliability and backup:** Cloud services can provide data backup, recovery and continuous availability.
5. **Easy collaboration and updates:** Multiple users can collaborate, while applications can be updated centrally.

For example, Google Drive allows users to store, access and share files through the cloud.

Section C — Detailed Solutions Including OR Parts

Q.3: Half-Duplex and Full-Duplex Modes

Half-Duplex Mode

In **half-duplex communication**, data can travel in both directions, but **only in one direction at a time**. When one device is transmitting, the other device must wait.

Example: Walkie-talkies. One person speaks while the other listens. They cannot speak simultaneously.

Full-Duplex Mode

In **full-duplex communication**, data can travel in both directions **simultaneously**. Both devices can send and receive data at the same time.

Example: A telephone or mobile-phone call. Both people can speak and listen simultaneously.

Comparison

Half-duplex

Communication occurs in both directions, but one direction at a time.

A device either transmits or receives at a particular moment.

It is comparatively slower.

Example: Walkie-talkie.

Full duplex

Communication occurs in both directions simultaneously.

A device can transmit and receive at the same time.

It is faster and more efficient.

Example: Telephone call.

OR: Role of AI in Speech Recognition and Computer Vision

1. AI in Speech Recognition

Speech recognition is the ability of a computer system to identify spoken language and convert it into text or commands.

AI plays the following roles:

1. **Speech-to-text conversion:** AI analyses audio signals and converts spoken words into written text.
2. **Voice-command recognition:** It enables users to control devices and applications through voice commands.
3. **Language processing:** AI identifies words, accents and speech patterns to understand the meaning of spoken sentences.
4. **Improved accuracy:** Machine-learning models learn from large speech datasets and become more accurate over time.

Examples: Google Assistant, Siri, voice typing and automated customer-support systems.

2. AI in Computer Vision

Computer vision is the ability of a computer to obtain and understand information from images and videos.

AI plays the following roles:

1. **Object recognition:** AI identifies objects such as vehicles, people, animals and road signs in images.
2. **Face recognition:** It identifies or verifies individuals by analysing their facial features.
3. **Image classification:** AI places images into categories based on their content.
4. **Visual decision-making:** It helps machines make decisions based on visual information.

Examples: Facial unlocking on smartphones, medical-image analysis, security-camera monitoring and object detection in self-driving vehicles.

Q.4: Flowchart to Find the Largest of Three Unequal Numbers

Flowchart 2: Flowchart to find the largest of three unequal numbers

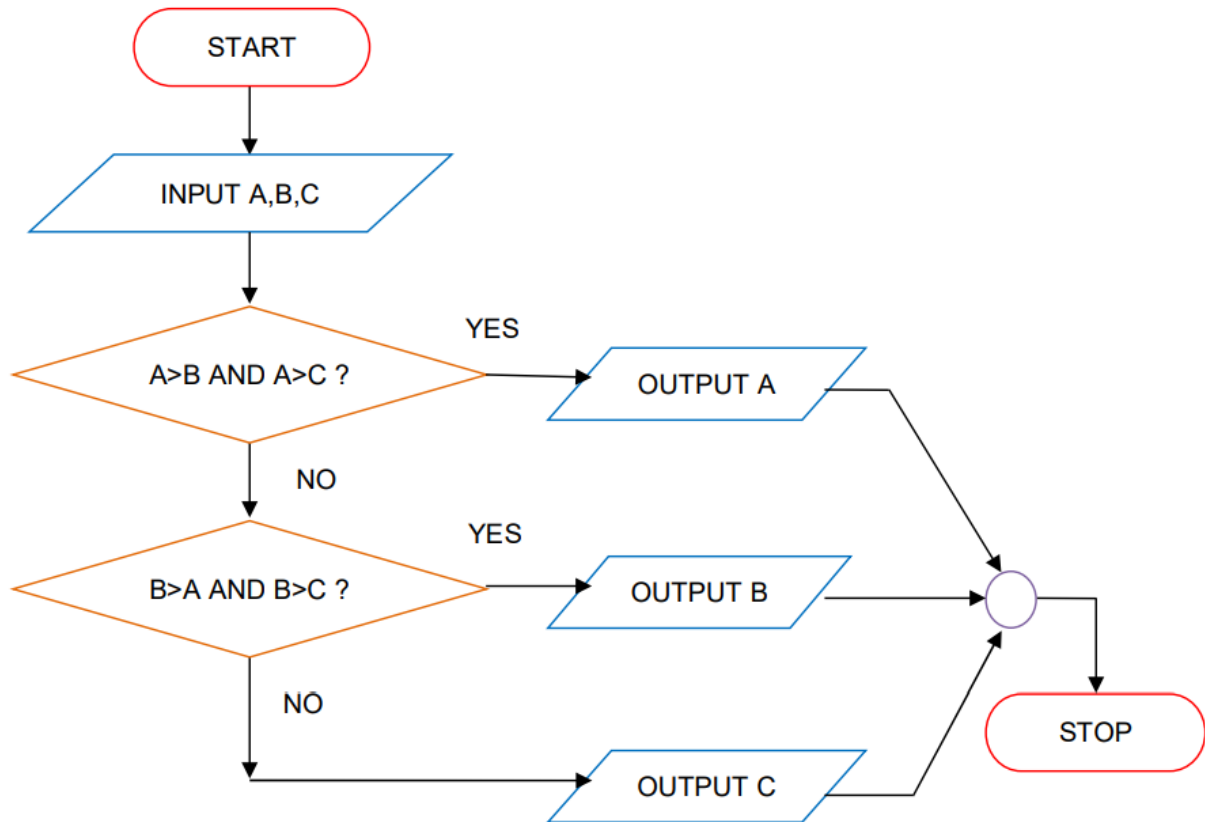


Fig.1-2 Flowchart to find largest of three numbers

OR: JavaScript Code to Calculate and Display the Sum of Two Variables

```
<!DOCTYPE html>
<html>
<head>
  <title>Sum of Two Numbers</title>
</head>

<body>
  <script>
    let number1 = 20;
    let number2 = 30;

    let sum = number1 + number2;

    document.write("The sum is: " + sum);
  </script>
</body>
</html>
```

A shorter JavaScript-only answer is:

 JavaScript 

```
let number1 = 20;
let number2 = 30;
let sum = number1 + number2;

console.log("The sum is: " + sum);
```

Q.5: Six Benefits of Computing Innovation

Computing innovation refers to a new or improved product, service or method that uses computing technology to solve problems or perform tasks more effectively.

1. Faster communication

Computing innovations allow people to communicate instantly through emails, messaging applications, social media and video conferencing.

Example: Students can attend an online class through Zoom or Google Meet.

2. Improved education

Digital learning platforms, simulations, educational applications and online libraries make learning more accessible, interactive and flexible.

Example: A student can watch recorded lectures and take online quizzes.

3. Better healthcare

Computers help doctors maintain patient records, diagnose diseases, analyse medical images and provide remote medical consultation.

Example: AI can assist doctors in detecting diseases from X-rays.

4. Increased productivity and automation

Computing innovations automate repetitive tasks, reduce human effort and allow work to be completed quickly and accurately.

Example: Payroll software automatically calculates employees' salaries.

5. Easy access to information

Search engines, websites and digital libraries provide quick access to a large amount of information from almost anywhere.

Example: Students can use an online library to find research material.

6. Improved business and banking services

Computing technology enables online shopping, digital payments, internet banking and efficient management of business records.

Example: Customers can transfer money through a mobile-banking application without visiting a bank.

OR: Categorical (Qualitative) Data and Its Types

Definition

Categorical data, also called **qualitative data**, represents the qualities, characteristics or categories of people, objects or events. It describes **what type or category** something belongs to rather than measuring it numerically.

Categorical data may sometimes be represented by numbers, but those numbers work only as labels and cannot be used for meaningful mathematical calculations.

Examples: Gender, blood group, eye colour, educational level and customer-satisfaction category.

The two main types of categorical data are:

1. Nominal Data

Nominal data consists of categories that have **no natural order or ranking**. One category cannot be described as higher or lower than another.

Examples:

- Blood groups: A, B, AB and O
- Eye colours: Brown, blue, green and black
- Cities: Islamabad, Lahore, Karachi and Peshawar
- Operating systems: Windows, Linux, macOS and Android

If blood groups are coded as 1 = A, 2 = B, 3 = AB and 4 = O, the numbers are only labels. Blood group 4 is not greater than blood group 2.

Binary data is a special form of nominal data having only two categories.

Examples: Yes/No, present/absent and pass/fail.

2. Ordinal Data

Ordinal data consists of categories having a **meaningful order or ranking**. However, the exact difference between consecutive categories cannot be measured.

Examples:

- Satisfaction level: Dissatisfied, neutral, satisfied
- Education level: Primary, secondary, higher secondary, university
- Performance: Poor, average, good, excellent
- Class position: First, second and third

For example, “excellent” performance is higher than “good” performance, but the exact numerical difference between the two categories is not known.

Comparison of Nominal and Ordinal Data

Nominal data

Categories have no natural order.

It is used only for identification or classification.

Example: Blood group.

Categories can only be compared as same or different.

Ordinal data

Categories have a meaningful order.

It is used for classification and ranking.

Example: Satisfaction level.

Categories can be compared as higher or lower.

Paper D - Complete Solution

SECTION A - MCQs (12 marks)

Choose the option shown below. The reason is included for conceptual understanding.

1. Which strategy explains how a product will reach customers?

Correct Answer: A — Go-to-market strategy

Reason: A go-to-market strategy explains how a product will be introduced, promoted, priced and delivered to target customers.

Why other options are wrong:

- **B — Budget plan:** Describes expected income, expenses and financial requirements.
 - **C — Risk analysis:** Identifies possible risks and their effects on a business.
 - **D — Product testing:** Checks whether a product works properly and meets requirements.
-

2. Why is algorithmic thinking useful while solving a problem?

Correct Answer: B — It provides a clear step-by-step solution

Reason: An algorithm presents instructions in a logical and ordered sequence for solving a problem.

Why other options are wrong:

- **A — It removes the need for inputs:** Most problems require input data to produce a result.
 - **C — It focuses only on outputs:** An algorithm includes inputs, processing steps and outputs.
 - **D — It avoids decision-making:** Algorithms may include conditions and decisions, such as if-else.
-

3. What does the DOM tree represent in an HTML document?

Correct Answer: C — Structure of web page elements

Reason: The Document Object Model (DOM) represents an HTML document as a tree of elements. For example, <html>, <head>, <body>, headings and paragraphs are arranged in a hierarchical structure.

Why other options are wrong:

- **A — Program logic, steps and results:** This describes an algorithm or flowchart, not the DOM.
 - **B — Visual design only:** CSS controls visual design; the DOM represents structure.
 - **D — Server configuration strategy:** Server configuration is related to back-end and server management.
-

4. Which language is most suitable for creating static web pages?

Correct Answer: A — HTML

Reason: HTML is used to create the structure and fixed content of static webpages.

Why other options are wrong:

- **B — PHP:** PHP is mainly used for server-side programming and dynamic webpages.
 - **C — SQL:** SQL is used to create, access and manage databases.
 - **D — Python:** Python is a general-purpose programming language, not the basic markup language for webpage structure.
-

5. Which aspect of CSS can be changed using JavaScript?

Correct Answer: D — Font size and color

Reason: JavaScript can access HTML elements through the DOM and change their CSS properties, such as font size, color, background color and spacing.

Example:

```
document.getElementById("heading").style.color = "blue";  
document.getElementById("heading").style.fontSize = "30px";
```

Why other options are wrong:

- **A — Page content:** JavaScript can change page content, but this is not an aspect of CSS.
- **B — HTML structure:** HTML structure is controlled by HTML elements. JavaScript can modify it, but it is not a CSS property.
- **C — Database records:** Database records are managed through database systems and server-side programming, not CSS.

6. Why is the CPU called the brain of the computer?

Correct Answer: C — It controls and processes instructions

Reason: The CPU fetches, decodes and executes instructions. It controls computer operations and processes data, so it is called the brain of the computer.

Why other options are wrong:

- **A — It stores data permanently:** Permanent storage is provided by hard drives, SSDs and other storage devices.
- **B — It displays output:** The monitor displays output; the CPU processes the information.
- **D — It supplies power:** The power supply unit provides electrical power to the computer components.

7. Why is speech recognition considered an important area of AI?

Correct Answer: D — It allows computers to understand spoken language

Reason: Speech recognition uses AI to convert human speech into text or commands. Examples include voice assistants and voice typing.

Why other options are wrong:

- **A — It designs websites:** Website design is mainly related to HTML, CSS and web development.
- **B — It increases storage capacity:** Storage capacity depends on storage hardware and technology.
- **C — It stores audio files:** Storing audio files is a storage function, not speech recognition.

8. Which term refers to deceptive emails or messages used to steal personal information?

Correct Answer: B — Phishing

Reason: Phishing is a cyberattack in which fake emails or messages trick people into sharing passwords, bank details or other personal information.

Why other options are wrong:

- **A — Hacking:** Hacking means gaining unauthorized access to a computer system or network. Phishing is one method that may be used to obtain login details.

- **C — Shareware:** Shareware is software distributed for trial or limited use.
 - **D — Freeware:** Freeware is software available free of cost.
-

9. What is an entrepreneurial solution mainly designed to do?

Correct Answer: A — Address a problem with a business idea

Reason: An entrepreneurial solution uses creativity and innovation to solve a real-world problem through a product, service or business idea.

Why other options are wrong:

- **B — Increase exams:** Exams are related to education, not entrepreneurship.
 - **C — Reduce technology use and increase manual labor:** Entrepreneurial solutions often use technology to improve efficiency.
 - **D — Copy existing products:** Entrepreneurship focuses on creating value and solving problems, not simply copying products.
-

10. Why must inputs, processes and outputs be identified in a problem?

Correct Answer: B — To understand how a solution works

Reason: Identifying inputs, processes and outputs clarifies what data is entered, what steps are performed and what result is produced.

Example:

- **Input:** Marks of a student
- **Process:** Calculate total and average
- **Output:** Total marks and average

Why other options are wrong:

- **A — To increase speed:** Speed may improve indirectly, but this is not the main purpose.
 - **C — To reduce storage:** Identifying IPO does not reduce storage requirements.
 - **D — To design graphics:** Graphics design is unrelated to the IPO model.
-

11. Why are computer ethics important?

Correct Answer: D — To guide responsible use

Reason: Computer ethics provide principles for using technology honestly, safely, respectfully and responsibly. They address privacy, copyright, data security and cyberbullying.

Why other options are wrong:

- **A — To increase speed:** Computer speed depends on hardware and software performance.
 - **B — To store data:** Data storage is performed by storage devices and databases.
 - **C — To design software:** Software design is a technical development activity, not the purpose of computer ethics.
-

12. Why is categorical data different from numerical data?

Correct Answer: C — It represents categories or labels

Reason: Categorical data describes groups or qualities using names or labels, such as colour, gender, blood group or type of operating system.

Why other options are wrong:

- **A — It is measured in units:** Data measured in units, such as height in centimetres or weight in kilograms, is numerical data.
- **B — It always changes:** Both categorical and numerical data can change.
- **D — It is continuous:** Continuous data is numerical and can take values within a range, such as height or temperature.

SECTION B - Short Questions (30 marks)

i. What is the role of the operating system in a computer system? Elaborate briefly.

Answer: The operating system is system software that manages hardware and software resources. It manages CPU processes, memory, files and input/output devices, provides a user interface, controls security and allows application software to run.

OR What is the importance of flowchart in solving a problem?

Answer: A flowchart is a graphical representation of an algorithm. It shows the sequence of actions and decisions using standard symbols, makes logic easy to understand, helps detect errors and provides a blueprint for coding.

ii. How logical thinking helps in solving a computational problem? Explain briefly.

Answer: Logical thinking uses facts, conditions and a clear sequence of reasoning. It helps identify the problem, select relevant information, make correct decisions and produce an algorithm that gives reliable results.

OR Differentiate between a static and a dynamic website.

Answer: A static website shows fixed content and usually changes only when its source files are edited. A dynamic website generates or updates content in response to users, databases or server processing, such as an online shopping site.

iii. Highlight the significance of application software. Give one example.

Answer: Application software performs tasks for users, such as writing, calculation, designing or communication. It improves productivity and meets specific user needs. Example: Microsoft Word is used to create documents.

OR What is meant by spyware? Give one example.

Answer: Spyware is malicious software that secretly monitors a user and collects information such as keystrokes or browsing activity. Example: a keylogger that records usernames and passwords.

iv. Briefly describe quantitative data. Give one example.

Answer: Quantitative data is numerical data that can be counted or measured and used in calculations. Example: a student scored 82 marks, or the temperature is 30°C.

OR A student performs calculations and logical comparisons in a program. Which part of the CPU carries out these operations? Explain briefly.

Answer: The Arithmetic Logic Unit (ALU) performs arithmetic calculations and logical comparisons. It receives data from registers, carries out the required operation and sends the result to a register or memory.

v. A news story spreads rapidly online without a reliable source. How can one identify whether it is fake news or credible?

Answer: Check the original source and author, publication date, supporting evidence and trustworthy references. Compare the claim with reputable news outlets or official websites, inspect the URL and headline, and do not share it until verified.

OR Compare Primary and Secondary data collection methods.

Answer: Primary data is collected first-hand for a specific study, e.g., interviews, surveys or experiments. Secondary data has already been collected by someone else and is obtained from books, reports, websites or government records. Primary data is more specific; secondary data is usually faster and cheaper.

vi. Correct any three errors in the given HTML code.

Correct HTML Code

```
<html>
<head>
  <title>Heading</title>
</head>
<body>
  <h1>Welcome to My Website</h1>
</body>
</html>
```

OR Briefly elaborate two features of star topology. Also draw diagram.

Answer: In star topology, every device has a separate link to a central hub or switch. It is easy to add, remove and troubleshoot devices. Failure of one cable affects only one device, but failure of the central device affects the whole network.

vii. How does Data Science differ from Data Analytics?

Answer: Data Science includes data collection, cleaning, programming, modelling and prediction. Data Analytics focuses mainly on examining existing data to identify patterns and support decisions.\

OR Write down HTML code to display ordered list: Memory; RAM; ROM.

```
<!DOCTYPE html>
<html>
<head>
  <title>Memory List</title>
</head>

<body>
  <u>Memory</u>

  <ol>
    <li>RAM</li>
    <li>ROM</li>
  </ol>
</body>
</html>
```

viii. Briefly explain the importance of data types in JavaScript. Give one example.

Answer: A data type tells JavaScript what kind of value a variable stores and which operations are valid. It helps the interpreter handle data correctly. Example: let age = 15; stores a number, while let name = "Ali"; stores text.

OR Why is the Internet called a network of networks? Explain briefly.

Answer: The Internet connects many independent networks - home, school, business and national networks – using common protocols such as TCP/IP. Therefore, it is a worldwide network made from interconnected networks.

ix. Why is computational thinking important in computer science?

Answer: It helps represent a problem logically, divide it into manageable parts, identify patterns, remove irrelevant details and design an algorithm that can be implemented and tested.

OR Briefly explain one ethical issue related to the use of AI in society.

Answer: Privacy is an ethical issue. AI systems may collect and analyze personal data without meaningful consent. The data can be misused, leaked or used for surveillance; therefore consent, security and responsible data handling are necessary.

x. List and briefly explain any three stages of the design thinking process.

Answer: Empathize: understand users and their needs. Define: clearly state the problem. Ideate: generate many possible solutions. Prototype: create a simple model. Test: evaluate it with users and improve it. Any three of these with explanations are correct.

OR What is meant by machine learning? Identify its two applications.

Answer: Machine learning is a branch of AI in which computers learn patterns from data and improve performance without being explicitly programmed for every case. Applications include spam-email filtering, recommendation systems, speech recognition, medical diagnosis and fraud detection.

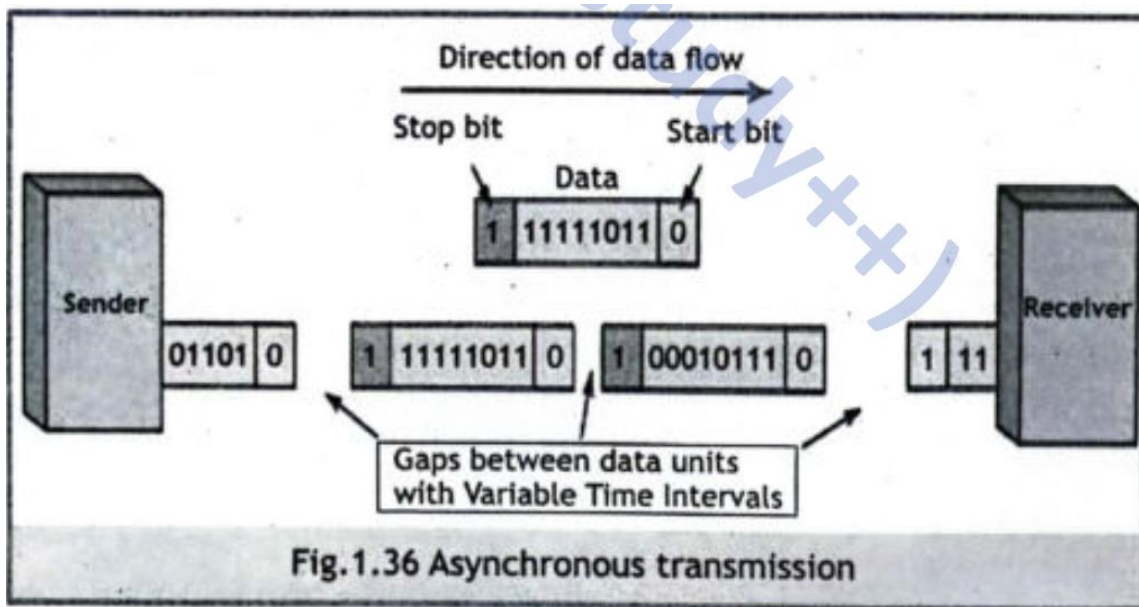
SECTION C — Complete Solutions Including OR Parts

Q.3: Compare Asynchronous and Synchronous Transmission Modes. Draw diagrams. (3+3)

Asynchronous Transmission

In asynchronous transmission, data is transmitted **one character or byte at a time**. Each character is enclosed between a **start bit** and one or more **stop bits**. The sender and receiver do not use a common clock. There may be an irregular time gap between two characters.

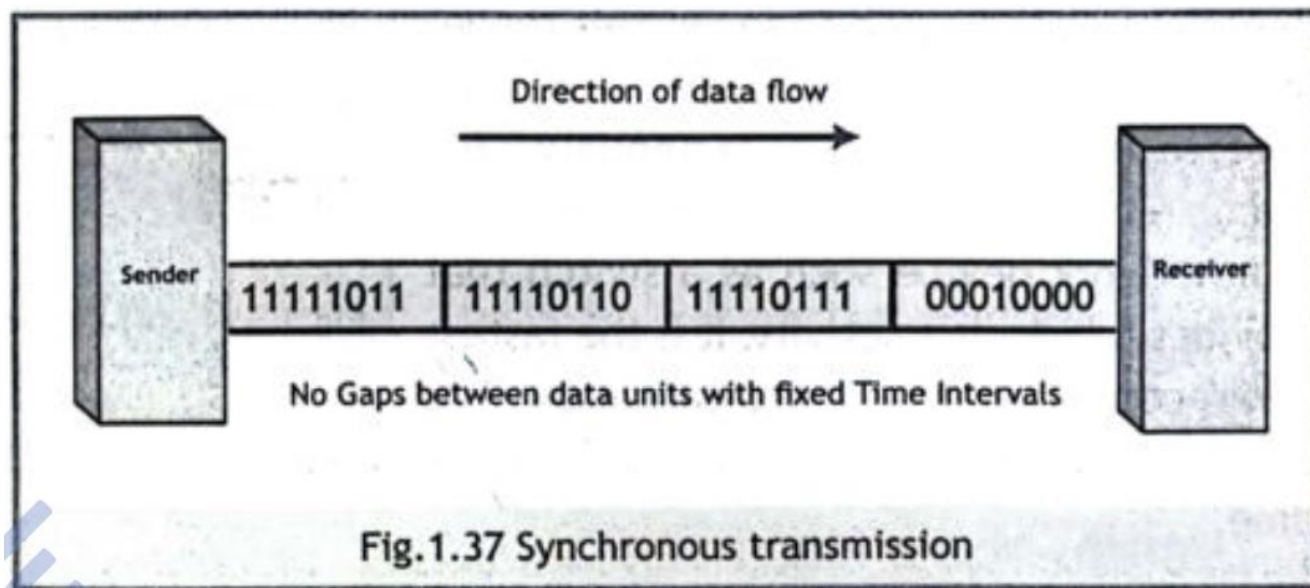
Diagram:



Synchronous Transmission

In synchronous transmission, data is transmitted as a **continuous stream in blocks or frames**. Individual characters do not require separate start and stop bits. The sender and receiver remain synchronized using timing or clock information.

Diagram:



Comparison

Basis	Asynchronous transmission	Synchronous transmission
Transmission method	Sends one character at a time	Sends data in blocks or frames
Synchronization	No continuously shared timing	Sender and receiver remain synchronized
Start/stop bits	Required for each character	Not required for every character
Speed	Relatively slow	Faster
Efficiency	Less efficient because of additional bits	More efficient for large amounts of data
Complexity	Simple and less expensive	More complex and relatively expensive
Suitable for	Small or irregular data transmission	Continuous and large data transmission
Example	Keyboard input, serial communication	Network and high-speed data communication

OR: How is AI Used in Healthcare and Education? Explain in Detail. (3+3)

A. Uses of AI in Healthcare

Artificial Intelligence helps healthcare professionals analyse medical information, diagnose diseases and improve patient care.

1. Disease diagnosis:

AI analyses X-rays, CT scans, MRI scans and laboratory reports to help doctors detect diseases such as cancer, pneumonia and heart disease.

2. **Personalized treatment:**

AI studies a patient's medical history, symptoms and test results to suggest treatment suitable for that patient.

3. **Patient monitoring:**

Smart devices can continuously monitor heartbeat, blood pressure and blood glucose levels. AI can warn doctors if an abnormal condition is detected.

4. **Drug discovery:**

AI analyses large amounts of scientific and medical data to identify possible medicines and speed up the development of new drugs.

5. **Virtual health assistants:**

AI chatbots can provide basic health information, remind patients to take medicine and help them book appointments.

Example: An AI system can examine a chest X-ray and highlight a suspicious area for further examination by a doctor.

AI assists healthcare professionals; important medical decisions should still be confirmed by qualified doctors.

B. Uses of AI in Education

AI helps improve teaching and learning by providing personalized and timely support.

1. **Personalized learning:**

AI can adjust lesson difficulty according to each student's performance and learning speed.

2. **Intelligent tutoring:**

AI-based tutors can explain concepts, provide examples and give immediate feedback to students.

3. **Automated assessment:**

AI can check objective tests and assignments, calculate results and identify common student mistakes.

4. **Learning recommendations:**

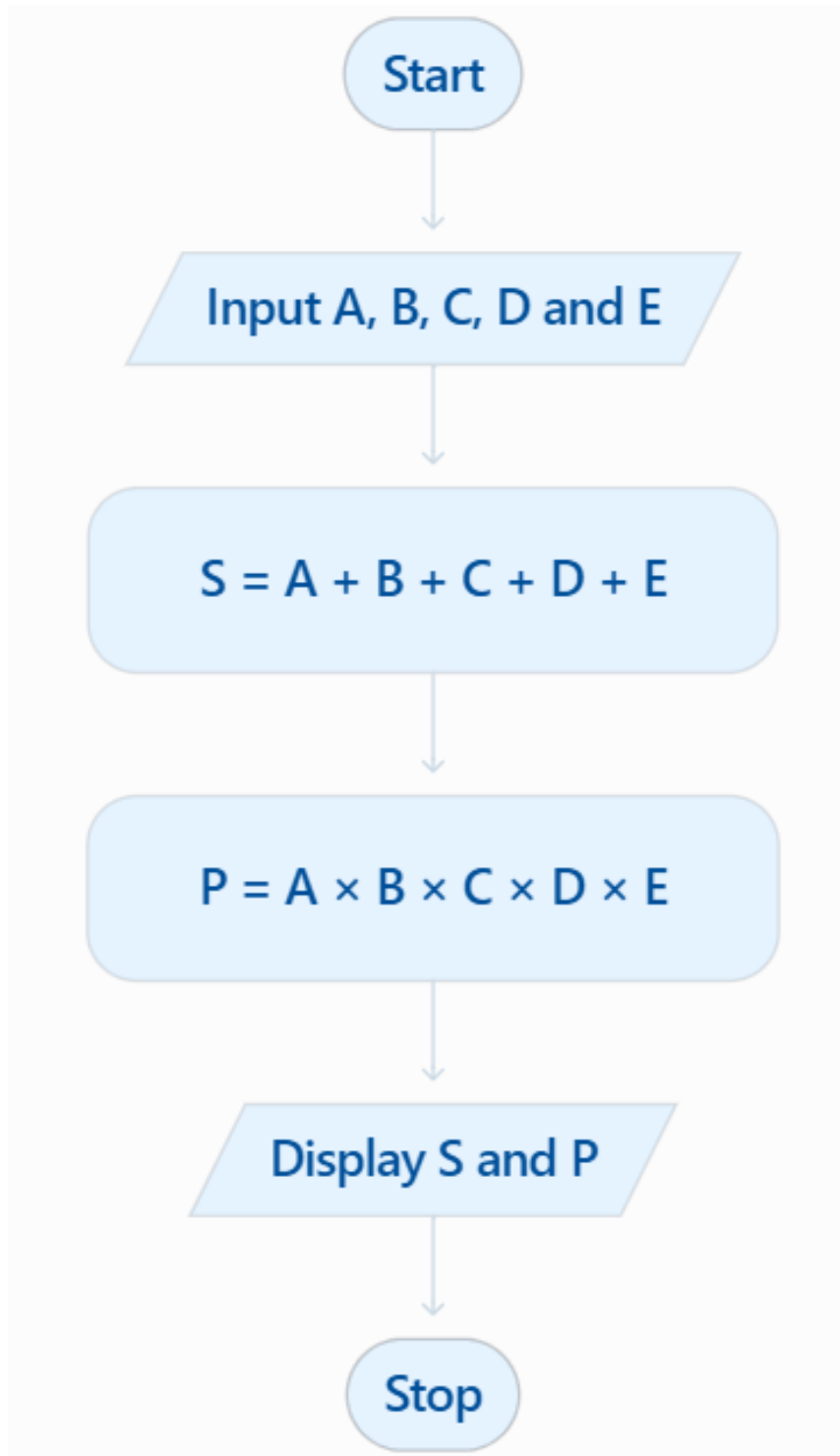
AI recommends suitable videos, exercises and reading material based on a student's strengths and weaknesses.

5. **Administrative support:**

Educational institutions can use AI for attendance management, timetable preparation and answering routine queries.

Example: If a student repeatedly makes mistakes in algebra, an AI learning system can provide additional algebra exercises and simpler explanations.

Q.4: Draw a Flowchart to Find the Sum and Product of Five Numbers. (6 marks)



OR: Write JavaScript Code to Display Numbers from 0 to 9. (6 marks)

Complete HTML and JavaScript program

```
<!DOCTYPE html>
<html>
<head>
  <title>Display Numbers</title>
</head>
<body>
  <h2>Numbers from 0 to 9</h2>

  <script>
    for (let i = 0; i <= 9; i++) {
      document.write(i + "<br>");
    }
  </script>
</body>
</html>
```

Q.5: Explain the Importance of the Following in the Safe and Responsible Use of Computers. (2×3)

(a) Keeping the Computer System Updated

Operating systems, browsers, antivirus programs and other applications should be updated regularly.

- Updates fix security weaknesses that hackers or malicious software may exploit.
- They protect the computer against newly discovered viruses and cyberattacks.
- Updates also correct software errors and improve system stability, compatibility and performance.

Example: Installing the latest security update can prevent malware from using a known weakness in the operating system.

(b) Using Strong and Unique Passwords

A strong password is long and contains an unpredictable combination of uppercase and lowercase letters, numbers and symbols.

- A strong password is difficult for attackers to guess or crack.
- A unique password prevents multiple accounts from being compromised if the password of one account is exposed.
- Users should not share passwords and should enable two-factor authentication where possible.

Example: A password such as F@khar!26Study# is stronger than 123456 or password.
Students should not copy this example as their real password.

(c) Backing Up Data

A backup is an additional copy of important data stored in a separate and secure location.

- It allows lost or damaged files to be recovered.
- It protects against hardware failure, accidental deletion, theft, viruses and ransomware attacks.
- Backups may be stored on an external drive or a trusted cloud storage service.
- Backups should be created regularly and tested to ensure that the files can be restored.

Example: If a computer's hard drive fails, documents can be recovered from a cloud or external-drive backup.

OR: What is Big Data? Explain Any Two Applications of Big Data in Detail. (2+4)

Definition of Big Data

Big Data refers to extremely large, rapidly generated and diverse collections of data that cannot be efficiently stored, processed and analysed using traditional data-processing methods.

Big Data is commonly described through the following characteristics:

- **Volume:** A very large quantity of data
- **Velocity:** Data is produced and processed at high speed
- **Variety:** Data exists in different forms, such as text, images, audio and video
- **Veracity:** The reliability and accuracy of data
- **Value:** Useful information obtained from the data

Application 1: Healthcare

Healthcare organizations generate data from patient records, laboratory reports, medical images, wearable devices and hospital systems. Big Data technology combines and analyses this information to:

- identify patterns related to diseases;
- support early diagnosis;
- predict disease outbreaks;
- monitor patients' health;

- improve treatment planning; and
- help researchers develop new medicines.

Example: Data collected from thousands of patients can be analysed to determine which treatment is most effective for a particular disease or group of patients.

Application 2: Business and E-Commerce

Businesses collect large amounts of data from customer purchases, website visits, product searches, reviews and social-media activity. This data is analysed to:

- understand customer preferences;
- recommend suitable products;
- forecast future demand;
- manage stock and supply chains;
- improve marketing campaigns; and
- detect suspicious transactions or fraud.

Example: An online shopping website analyses a customer's searches and previous purchases to recommend products that the customer may be interested in buying.

Paper M - Complete Solution

SECTION A - MCQs (12 marks)

Choose the option shown below. The reason is included for conceptual understanding.

1. Which stage of design thinking focuses on understanding users' needs?

Correct Answer: B — Empathizing

Reason: The empathizing stage involves understanding users' feelings, problems, needs and experiences.

Why other options are wrong:

- **A — Ideating:** Generates possible ideas and solutions.
 - **C — Prototyping:** Creates an early model of a proposed solution.
 - **D — Testing:** Evaluates the prototype with users to identify improvements.
-

2. Why is algorithmic thinking useful while solving a problem?

Correct Answer: A — It provides a step-by-step solution

Reason: Algorithmic thinking organizes instructions in a logical sequence to solve a problem correctly.

Why other options are wrong:

- **B — It removes decision-making:** Algorithms may include decisions using conditions such as if-else.
 - **C — It focuses only on outputs:** An algorithm includes inputs, processing and outputs.
 - **D — It avoids planning:** Designing an algorithm requires proper planning and logical thinking.
-

3. Why should users protect personal data online?

Correct Answer: A — To prevent misuse

Reason: Personal data such as passwords, CNIC numbers, bank details and photographs can be misused for fraud, identity theft or unauthorized access.

Why other options are wrong:

- **B — To share faster:** Protecting data means controlling access, not sharing it more quickly.
 - **C — To reduce cost:** Data protection is mainly related to privacy and security, not cost reduction.
 - **D — To improve graphics:** Graphics are unrelated to personal-data protection.
-

4. Which of the following is an example of a secondary data collection method?

Correct Answer: D — Government record

Reason: Government records have already been collected and prepared by a government organization. Using such existing information is secondary data collection.

Why other options are wrong:

- **A — Experiment:** Data is collected directly by the researcher, so it is primary data.
 - **B — Observation:** The researcher directly observes and records information, making it primary data.
 - **C — Interviews:** Information is collected directly from people, so interviews are a primary data method.
-

5. How does a flowchart help in solving a problem?

Correct Answer: C — It visually shows the sequence of steps

Reason: A flowchart uses symbols and arrows to display the logical sequence of operations and decisions in an algorithm.

Why other options are wrong:

- **A — It stores data:** Data is stored in memory, databases or storage devices.
 - **B — It executes the program:** A flowchart only represents an algorithm; it does not execute code.
 - **D — It replaces algorithms and performs extra calculations:** A flowchart is a visual representation of an algorithm and does not perform calculations itself.
-

6. Why is identifying inputs, processes and outputs important in a problem?

Correct Answer: B — To understand how a solution works

Reason: The IPO model clearly explains the data entered, the operations performed and the result produced.

Example:

- **Input:** Two numbers
- **Process:** Add the numbers
- **Output:** Their sum

Why other options are wrong:

- **A — To reduce memory usage:** Identifying IPO does not directly reduce memory requirements.
- **C — To speed up execution:** It may help design a better solution, but increasing speed is not its main purpose.
- **D — To create diagrams for visualization of the program:** Flowcharts are used for visual diagrams; IPO identification mainly explains how a system works.

7. Why are hyperlinks important in HTML documents?

Correct Answer: A — They connect one page to another

Reason: Hyperlinks allow users to move from one webpage, document or section to another.

Example:

```
<a href="https://example.com">Visit Website</a>
```

Why other options are wrong:

- **B — They change page style:** CSS changes page style and appearance.
- **C — They store images:** Images are inserted using the `<img>` tag.
- **D — They run scripts:** JavaScript is used to run scripts.

8. Web development is mainly concerned with:

Correct Answer: D — Creating websites and web applications

Reason: Web development involves designing, coding, testing and maintaining websites and web applications.

Why other options are wrong:

- **A — Computer hardware:** Hardware deals with physical computer components.

- **B — Managing networks:** Network administration deals with connecting and managing networks.
 - **C — Installing software:** Software installation is a general computer-maintenance task, not the main purpose of web development.
-

9. Why are arrays useful in JavaScript?

Correct Answer: B — To store multiple values in a single variable

Reason: An array stores several related values under one variable name.

Example:

```
let fruits = ["Apple", "Mango", "Banana"];
```

Why other options are wrong:

- **A — To store one value only:** A normal variable can store one value; an array stores multiple values.
 - **C — To change page style and color:** CSS and JavaScript style properties are used for this purpose.
 - **D — To link web pages:** HTML hyperlinks using the <a> tag connect webpages.
-

10. Why is a computer considered an artificial system?

Correct Answer: C — It is man-made to perform tasks

Reason: A computer is designed and developed by humans to perform calculations, process data and execute instructions.

Why other options are wrong:

- **A — It grows naturally:** Computers do not grow or develop naturally.
 - **B — It exists in nature:** Computers are human-made devices, not natural systems.
 - **D — It works without instructions:** A computer requires instructions or programs to perform tasks.
-

11. Why is Artificial Intelligence (AI) useful in solving real-world problems?

Correct Answer: D — It enables machines to make intelligent decisions

Reason: AI allows machines to analyse data, recognize patterns, learn from experience and make decisions or predictions.

Why other options are wrong:

- **A — It stores large files:** Data storage is performed by storage devices and databases.
 - **B — It improves hardware speed:** Hardware speed depends on processors and other physical components.
 - **C — It replaces the Internet:** AI uses the Internet in many applications but does not replace it.
-

12. Which issue arises due to the misuse of personal data online?

Correct Answer: C — Privacy violation

Reason: Misusing personal data can expose private information, cause identity theft, enable unauthorized tracking or allow access to personal accounts.

Why other options are wrong:

- **A — Increased speed:** Misuse of data does not increase computer or Internet speed.
- **B — Improved usability:** Misuse creates security and privacy risks, not improved usability.
- **D — Better communication:** Misuse of personal data can harm users; it does not improve communication.

SECTION B - Short Questions (30 marks)

i. Write down the output of the following JavaScript code: `reward=5555; a=reward%10; b=reward/10.`

Answer: a = 5 because 5555 divided by 10 leaves remainder 5. b = 555.5 because ordinary division is used. Therefore, the output is: **5....555.5.**

OR Why is packet switching efficient for data transmission? Explain briefly.

Answer: It divides data into small packets and shares network links efficiently. Packets can take available routes, so one busy or failed route does not stop the whole message. It supports many users and retransmits only lost packets.

ii. How logical thinking helps in solving a computing problem? Explain briefly.

Answer: It uses facts, conditions and a step-by-step sequence to reach a valid result. It helps select relevant information, compare alternatives, detect errors and design a correct algorithm.

OR What is cloud computing? Why is it needed in modern computing environment?

Answer: Cloud computing is the delivery of storage, software, processing and other services over the Internet. It is needed for access from anywhere, collaboration, scalable resources, lower hardware cost, backup and quick

deployment.

iii. Enlist any six Sustainable Development Goals (SDGs).

Answer: Any six: No Poverty; Zero Hunger; Good Health and Well-being; Quality Education; Gender Equality; Clean Water and Sanitation. Other official SDGs are also acceptable.

OR How do machine learning help computers solve problems? Give examples of its applications.

Answer: Machine learning trains a model on data so it can identify patterns, predict outcomes or classify new cases. Examples include face recognition, spam filtering, speech recognition, recommendation systems and disease

prediction.

iv. What is meant by output device? Write names of any three output devices.

Answer: An output device presents processed information from a computer to the user. Examples: monitor, printer, speakers, projector and headphones.

OR What is meant by software piracy? Give one example.

Answer: Software piracy is unauthorized copying, installation, distribution or use of copyrighted software. Example: installing a cracked copy of paid software or sharing its license key without permission.

v. What is a concept map? How does it represent knowledge?

Answer: A concept map is a graphical representation of concepts and relationships. Concepts are placed in nodes or boxes and connected with labelled lines or arrows, showing hierarchy and links between ideas.

OR Give one use of hyperlinks in HTML documents. Also write syntax for defining a link.

Answer: A hyperlink connects a web page to another page, file, section or website.

Syntax: `<a href="https://example.com">Visit website</a>`.

vi. How is data transferred in synchronous transmission? Also draw diagram.

Answer: Data is transferred as a continuous stream or block. Sender and receiver use synchronized timing, and control information identifies the beginning and end of the frame. It is efficient for large blocks of data.

OR Write down the purpose of any two flowchart symbols.

Answer: (1) Terminator (oval): shows the Start or Stop of an algorithm.

(2) Process (rectangle): represents an operation or calculation, such as $SUM = A + B$.

vii. Briefly describe Volume, Velocity and Variety in Big Data.

Answer: Volume is the huge amount of data, such as billions of social-media records. Velocity is the speed at which data is generated and processed, such as live sensor data. Variety means different forms of data, such as text, images, audio, video and structured tables.

OR How Software engineering differs from Hardware engineering?

Answer: Software engineering designs, develops, tests and maintains programs and systems. Hardware engineering designs and tests physical components such as processors, circuits, memory and devices.

viii. A user receives an email asking for bank details with an urgent warning. Identify the type of cyber threat and justify your answer.

Answer: It is phishing. The message uses deception, urgency and impersonation to make the user disclose sensitive bank information through email or a fake link. The user should not click, reply or share details; it should be reported and deleted.

OR How is data science helpful in solving business problems?

Answer: Data science combines statistics, programming and machine learning to analyse business data. It helps forecast sales, understand customers, detect fraud, optimize operations and support evidence-based decisions.

ix. Briefly explain the role of front-end development in a web project.

Answer: Front-end development creates the visible and interactive part of a website that runs in the user's browser. HTML provides structure, CSS provides appearance and JavaScript adds behavior and interaction.

OR Why is the Internet considered a global network of networks?

Answer: It connects millions of independent networks around the world through common protocols. Devices and networks can communicate even when they belong to different organizations or countries.

x. Briefly discuss primary data collection methods. Also identify any two examples.

Answer: Primary data is collected directly from original sources for a specific purpose. Methods include surveys/questionnaires, interviews, observations, experiments and focus groups. Two examples are interviewing students about study habits and observing traffic at a road junction.

OR Write HTML code to display an unordered list: Software; System Software; Application Software.

```
<html>
<head>
  <title>Types of Software</title>
</head>
<body>

  <ul>
    <li><u>Software</u>
      <ol>
        <li>System Software</li>
        <li>Application Software</li>
      </ol>
    </li>
  </ul>

</body>
</html>
```

SECTION C — Complete Solutions Including OR Parts

Q.3: Importance in the Safe and Responsible Use of Computers (2 × 3 = 6)

(a) Backing Up Data

A **backup** is an additional copy of important files stored in a separate, secure location.

- It protects data against accidental deletion, hardware failure, theft, viruses and ransomware.
- It allows users to restore their files if the original data is lost or damaged.
- Backups should be created regularly and stored on an external drive or trusted cloud service.

Example: If a computer's hard disk fails, documents can be recovered from a cloud backup.

(b) Use of Secure Wi-Fi

Secure Wi-Fi is a wireless network protected by strong encryption and a password.

- It prevents unauthorized people from accessing the network.
- It protects personal information, passwords and online activities from interception.
- Users should select networks protected by **WPA2 or WPA3** and avoid sensitive transactions on open public Wi-Fi.
- The default router password should be replaced with a strong and unique password.

Example: Using password-protected home Wi-Fi is safer than entering banking information through an open public network.

(c) Privacy Settings

Privacy settings allow users to control how their personal information is collected, used and shared.

- They can restrict who may view posts, photographs, location and contact information.
- They prevent applications from unnecessarily accessing the camera, microphone or location.
- Proper privacy settings reduce the risks of identity theft, cyberstalking and misuse of personal data.
- These settings should be reviewed regularly because applications may update their policies.

Example: A social-media account can be set to **Private** so that only approved people can view its content.

OR: What Is Big Data? Write Any Four Benefits of Big Data. (2 + 4 = 6)

Definition

Big Data refers to extremely large, complex and rapidly generated collections of data that cannot be stored, processed and analysed efficiently through traditional data-processing methods.

Big Data commonly has the following characteristics:

- **Volume:** enormous quantity of data
- **Velocity:** high speed of data generation
- **Variety:** different formats such as text, images, audio and video

Four Benefits of Big Data

1. **Better decision-making:**
Organizations analyse facts, trends and patterns to make faster and more accurate decisions.
2. **Improved customer services:**
Businesses study customers' searches, purchases and feedback to understand their needs and provide personalized services.
3. **Fraud and risk detection:**
Banks and other organizations analyse transactions to identify unusual activities and detect possible fraud.
4. **Cost and time reduction:**
Big Data helps organizations improve operations, reduce waste, forecast demand and use resources efficiently.

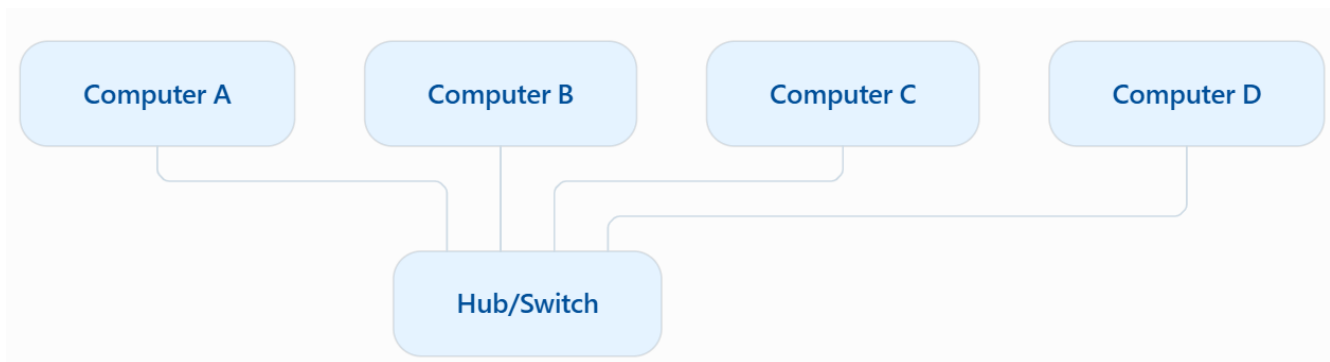
Other acceptable benefits: disease prediction, scientific research, traffic management and targeted marketing.

Q.4: How Does Star Topology Differ from Bus Topology? Explain and Draw Diagrams. (3 + 3 = 6)

Star Topology

In a **star topology**, every computer or device is connected through a separate cable to a central device, usually a **hub or switch**. All communication passes through the central device.

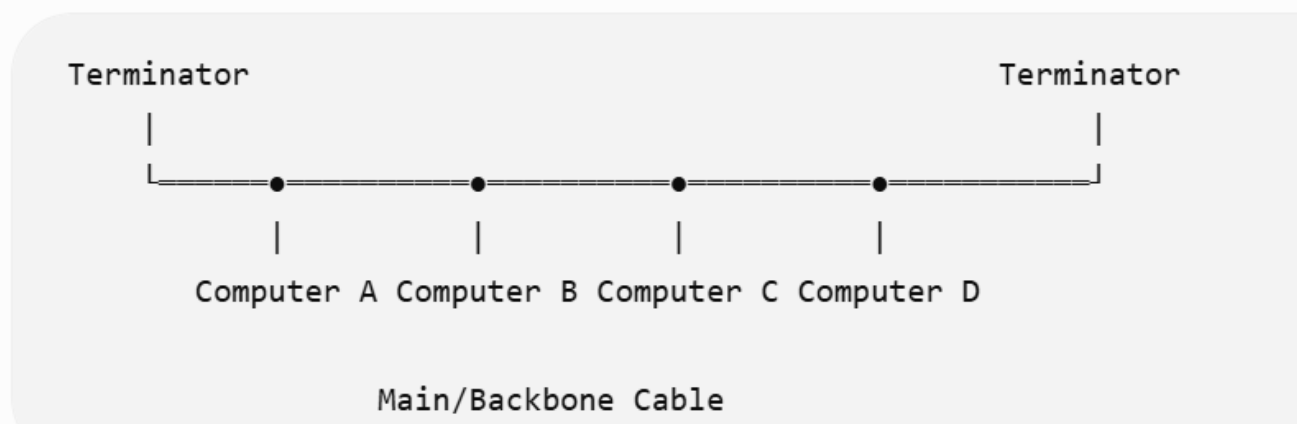
Diagram of Star Topology



Bus Topology

In a **bus topology**, all computers are connected to a single main cable called the **backbone cable**. Data travels along this shared cable. A **terminator** is installed at each end to prevent signals from reflecting.

Diagram of Bus Topology



Comparison

Basis	Star topology	Bus topology
Structure	Every device connects to a central hub or switch	Every device connects to one backbone cable
Cable requirement	Requires more cable	Requires less cable
Failure of one device cable	Only that particular device is disconnected	Usually does not affect other devices
Main device/cable failure	Failure of the hub or switch stops the whole network	Failure of the backbone stops the whole network
Performance	Generally better because devices have separate connections	Performance decreases when network traffic increases
Troubleshooting	Faults are easier to locate	Faults are more difficult to locate
Installation cost	Relatively expensive	Relatively inexpensive
Expansion	Devices can be added easily if ports are	Adding devices may disturb the

Basis

available

Star topology

Bus topology

network

Conclusion

Star topology is more reliable and easier to manage, while bus topology is simpler and cheaper for a small network.

OR: How Is Artificial Intelligence Used in Education and Gaming? (3 + 3 = 6)

A. Uses of AI in Education

Artificial Intelligence improves teaching and learning by analysing students' performance and providing appropriate assistance.

1. **Personalized learning:**
AI changes the difficulty and content of lessons according to a student's learning speed, strengths and weaknesses.
2. **Intelligent tutoring systems:**
AI tutors explain concepts, answer questions and provide immediate feedback.
3. **Automated assessment:**
AI can check quizzes and assignments, calculate results and identify common mistakes.
4. **Learning recommendations:**
It recommends suitable videos, exercises and reading material based on a student's performance.
5. **Administrative support:**
AI assists institutions in preparing timetables, maintaining attendance and answering routine queries.

Example: If a student repeatedly makes mistakes in algebra, an AI learning system can provide additional algebra exercises and simpler explanations.

B. Uses of AI in Gaming

AI makes games more realistic, challenging and responsive.

1. **Intelligent non-player characters:**
AI controls computer-operated characters and allows them to react to the player's actions.
2. **Difficulty adjustment:**
AI can automatically increase or decrease the difficulty according to the player's performance.
3. **Pathfinding:**
AI helps game characters find suitable routes and avoid obstacles.
4. **Content generation:**
AI can create maps, levels, missions and virtual environments.

5. Player-behaviour analysis:

AI analyses playing patterns to improve gameplay and detect cheating.

Example: In a racing game, AI controls opponent vehicles and adjusts their behavior according to the player's skill level.

Q.5: Online Order System—Inputs, Processes and Outputs (2 + 2 + 2 = 6)

An online order system follows the **Input–Process–Output (IPO)** cycle.

Stage	Details
Inputs	Customer's name and contact details; selected products and quantities; delivery address; payment information; discount code
Processes	Validate customer information; check product availability; calculate item cost, discount, tax and delivery charges; process payment; update inventory; generate an order number
Outputs	Order-confirmation message; receipt or invoice; order number; payment status; delivery details; updated stock record

Example

Suppose a customer orders two books costing Rs. 500 each.

Input:

- Product: Book
- Quantity: 2
- Price per book: Rs. 500
- Customer and delivery details

Processing:

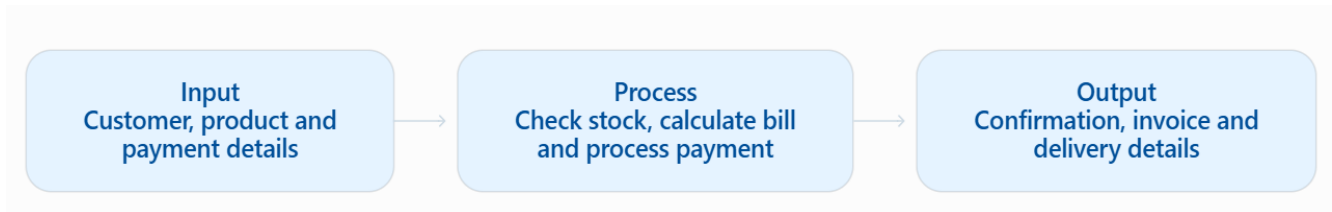
$$\begin{aligned}\text{Total cost} &= \text{Quantity} \times \text{Price} \\ \text{Total cost} &= 2 \times 500 = \text{Rs. } 1,000\end{aligned}$$

The system checks the stock, processes payment and updates the inventory.

Output:

Order confirmed
Total amount: Rs. 1,000
Order number: 1025
Payment status: Successful

IPO Diagram



OR: Write JavaScript Code to Display Numbers from 10 to 1. (6 marks)

Complete HTML and JavaScript Program

```
<html>
<head>
  <title>Numbers from 10 to 1</title>
</head>
<body>
  <h2>Numbers from 10 to 1</h2>

  <script>
    for (let i = 10; i >= 1; i--) {
      document.write(i + "<br>");
    }
  </script>
</body>
</html>
```